How do Organisms Reproduce?

The big idea

Reproduction makes more individuals of a species. The key event is **DNA copying**: the copies are almost, but not exactly, the same. These small **variations** help a population survive changes in the environment.

Asexual reproduction (one parent)

Mode	Example
Fission	Binary fission in Amoeba; multiple fission in Plasmodium
Fragmentation	Spirogyra breaks into pieces that grow
Regeneration	Planaria and Hydra regrow from pieces
Budding	Hydra, yeast
Vegetative propagation	Potato (stem), Bryophyllum (leaf buds), sugarcane, rose; by cutting, layering, grafting
Spore formation	Rhizopus (bread mould) spores in sporangia

Vegetative propagation gives plants that bear fruit sooner and keeps desired traits exactly.

Sexual reproduction in flowering plants

The flower is the reproductive part. **Stamens** (male) make pollen; the **pistil** (female) has stigma, style and ovary with ovules.

FROM POLLEN TO SEED

Pollination: transfer of pollen to the stigma (self or cross, by wind, water, insects).
Fertilisation: a pollen tube grows down the style; the male germ cell fuses with the female germ cell in the ovule to form a **zygote**.
Afterwards the ovule becomes the **seed** and the ovary becomes the **fruit**.
The seed has an embryo that grows into a new plant (germination).

Human reproduction

At **puberty** the body changes: in boys, voice deepens and facial hair grows; in girls, breasts develop and menstruation begins. These are controlled by hormones (testosterone, oestrogen).

- **Male**: testes make sperm and testosterone; sperm travel through the vas deferens, mixed with fluids from the seminal vesicles and prostate.
- **Female**: ovaries release one egg a month; it travels down the **oviduct** (where fertilisation happens) to the **uterus**. The embryo implants in the uterus wall and gets nourishment through the **placenta**. If there's no fertilisation, the uterus lining breaks down: **menstruation**.

Reproductive health

Sexually transmitted diseases include gonorrhoea, syphilis (bacterial) and HIV-AIDS (viral). Contraceptive methods are mechanical (condoms), chemical (oral pills), surgical (vasectomy, tubectomy) and copper-T. Female foeticide (sex-selective abortion) is illegal in India, and maintaining a healthy sex ratio is important.

Quick recap

- DNA copying creates variation, which helps species survive.

- Asexual: fission, fragmentation, regeneration, budding, vegetative propagation, spores.
- Plants: pollination → fertilisation → seed and fruit.
- Humans: fertilisation in the oviduct; embryo nourished via the placenta.

Practice questions

1. Planaria regrows a whole body from a piece. This is
   (a) budding (b) fission (c) regeneration (d) spore formation
2. Bryophyllum reproduces vegetatively through buds on its
   (a) stem (b) roots (c) leaves (d) flowers
3. After fertilisation, the ovary develops into the
   (a) seed (b) fruit (c) embryo (d) flower
4. In humans, fertilisation normally takes place in the
   (a) uterus (b) oviduct (fallopian tube) (c) ovary (d) vagina
5. Variation in a species is important because it
   (a) makes all offspring identical (b) helps the species survive changes in the environment (c) stops reproduction (d) prevents DNA copying

Answers: 1. c 2. c 3. b 4. b 5. b